



Category: AI & Agentic

Business Concepts: Responsible AI, Compliance, Security

AI Safety & Governance Layer

Team Name : PERCEPTRON

College Name : Panimalar Engineering College

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Team details

Team Name - PERCEPTRON

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PROBLEM STATEMENT DETAILING

- Enterprises deploying AI/agent systems risk leaking PII and sensitive data through prompts, tool calls, and responses, with no visibility into what was exposed.

Goal: give teams a real-time layer that catches and contains this exposure.

- **Sub-tasks:** (1) detect PII/sensitive entities in real time across prompts and responses,
(2) mask or redact before data leaves the boundary,
(3) generate tamper-evident audit logs.

Challenge: do this with low latency inside live agent pipelines.

- **Success measures:** PII detection accuracy (precision/recall), masking latency added per request, % of sensitive events captured in audit logs, false-positive rate on legitimate content.

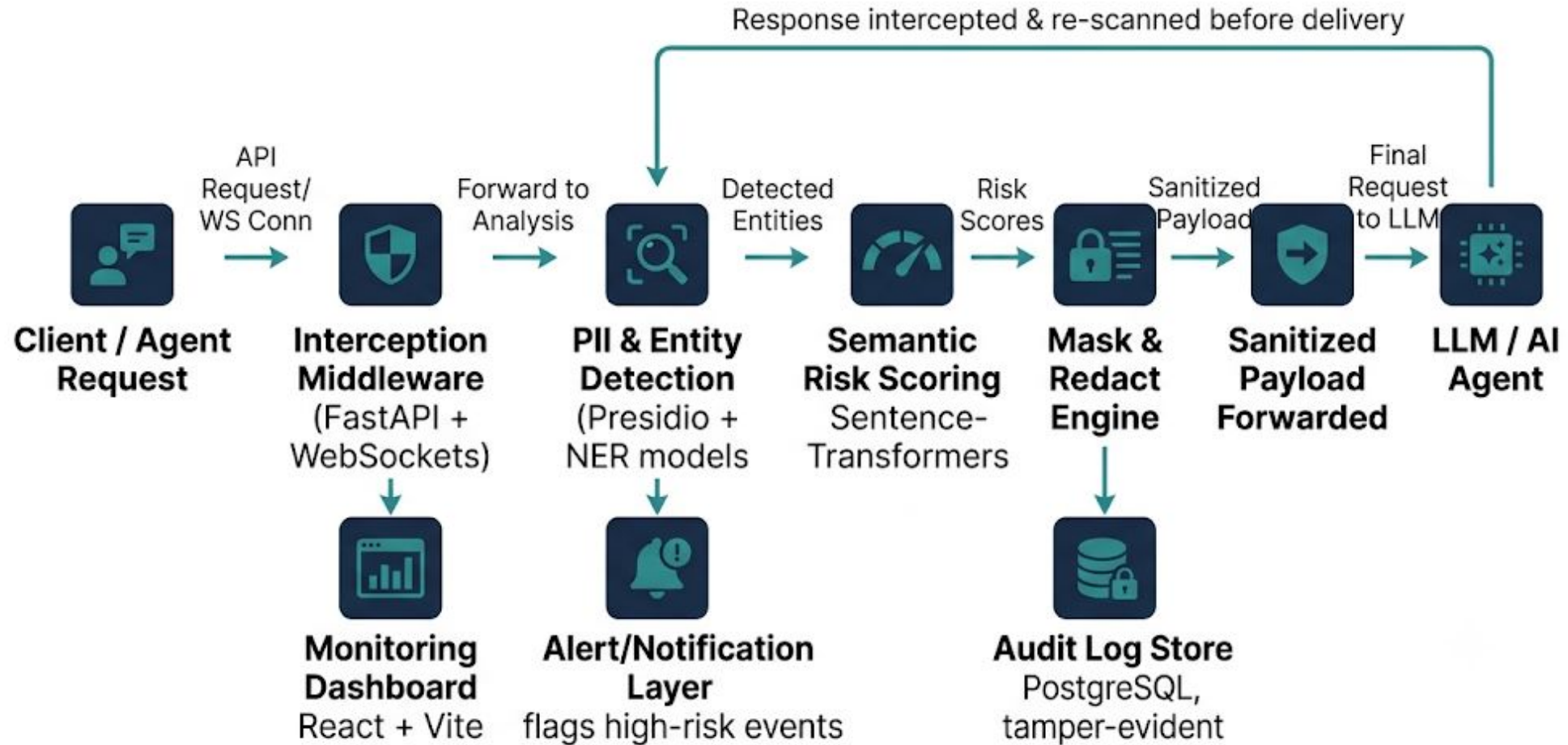
SOLUTION

- We propose an **AI Privacy Firewall** to protect users personal and sensitive information.
- It works as a **security layer between the user and the AI model**.
- It checks every **user prompt** for sensitive information like phone numbers, emails, Aadhaar numbers, names, and addresses.
- If sensitive information is found, it is **automatically masked** before sending the prompt to the AI.
- The AI's **response is also checked** for sensitive information before it reaches the user.
- All detected information and security actions are **stored in an audit log**.
- A **dashboard** allows administrators to monitor security events and risk levels.
- This helps **prevent personal data leaks while using AI systems**.

TECH STACKS

- **Backend:** FastAPI + WebSockets (real-time interception)
- **PII Detection:** Microsoft Presidio (spaCy NER) + Custom Regex Engine (8 patterns incl. Aadhaar, PAN)
- **Toxicity:** Detoxify (BERT-based, severity-stratified scoring)
- **Threat Detection:** Custom malware scanner (20 patterns: SQLi, XSS, PowerShell, reverse shells) + Advanced injection (polyglot, homoglyph, multilingual)
- **LLM Layer:** Multi-provider fallback chain
- **Audit:** SHA-256 hash-chained SQLite log (tamper-evident)
- **Dashboard:** Vanilla HTML/CSS/JS single-file SOC dashboard with WebSocket live flow
- **Performance:** TTL-based LRU cache, per-IP rate limiting, background ML loading

Architecture Diagram



CASE STUDY

- **Samsung, 2023** — engineers pasted proprietary source code into ChatGPT for debugging; data left the company boundary permanently. Samsung banned public GenAI tools internally afterward.
- **Slack AI indirect prompt injection, 2024** — a hidden instruction planted in a public channel message got followed later, exfiltrating private channel data.
- **MCP Tool Poisoning (Invariant Labs, 2025)** — malicious MCP server exfiltrated WhatsApp chat histories via poisoned tool metadata. One line each, framed as "this is the exact failure mode Sentinel is built to catch."

User Interface

Sentinel

AI GOVERNANCE LAYER

MONITOR

SOC · Live Flow

Live Overview

Event Stream2463

Audit Logs

Playground

AI Chat

CONFIGURE

Detection Rules

Connected Agents

Firewall active

5 agents · 4137 req/min

AI Chat — Safe Conversations

sentinel-prod-01 · us-east

STREAMING

presidio: ready

SR

AI Chat

Chat with AI through Sentinel's interception layer — safe prompts reach the LLM, PII/toxic prompts are blocked

You · 12:21:27

hi

Sentinel · 12:21:27

Scan: Clean · 4757ms

AI Assistant · 12:21:34

Hello, how can I assist you today?

You · 12:22:07

My email is priya.sharma@gmail.com, please update my records. Contact me at +91 98765 43210 for the interview. My Aadhaar number is 2345 6789 0123, please verify my KYC. PAN card number is BXPQK1234M for tax filing. Card number 4111 1111 1111 1111, please process the refund. API key sk_live_a1b2c3d4e5f6g7h8i9j0 needs to be rotated.

Sentinel · 12:22:07

Scan: Email, Gov ID (PAN), Aadhaar, Credit Card, Phone, API Key — BLOCKED · 163ms

Sentinel · 12:22:07

Blocked: Email (MASK), Gov ID (PAN) (BLOCK), Aadhaar (BLOCK), Credit Card (MASK), Phone (MASK), API Key (STRIP)

Your message was NOT sent to the AI.

Ask me anything... (try including an email or Aadhaar to see blocking)

Press Enter to send · Shift+Enter for new line

Interception Inspector

Last scan · 14:42:12

VERDICT

BLOCKED

ENTITIES (2)

Gov ID (PAN)0.98BLOCK

Aadhaar0.99BLOCK

TOTAL LATENCY

65ms

CLEANED (SENT TO AI)

I need to write a message to customer support regarding my KYC verification. My name is Priya Sharma and my PAN number is [GOV_ID]. My Aadhaar number is [AADHAAR]. Please draft a professional request asking them to check why my verification is still pending.

BLOCKED REASONS

Gov ID (PAN) (BLOCK), Aadhaar (BLOCK)

User Interface

 **Sentinel**
AI GOVERNANCE LAYER

MONITOR

 SOC · Live Flow

 Live Overview

 Event Stream 2481

 Audit Logs

 Playground

 **AI Chat**

CONFIGURE

 Detection Rules

 Connected Agents

Firewall active

5 agents · 4137 req/min

AI Chat — Safe Conversations

sentinel-prod-01 · us-east

STREAMING

presidio: ready

SR

AI Chat

Chat with AI through Sentinel's interception layer — safe prompts reach the LLM, PII/toxic prompts are blocked

You · 12:22:18

hi my email is xxx

Sentinel · 12:22:18

Scan: Clean · 2535ms

AI Assistant · 12:22:21

It looks like you only shared part of your email address. I'm happy to chat with you, but I want to make sure your email address remains private. Is there something specific you'd like to talk about or ask, or would you like some help with something else?

You · 14:23:55

Hi, can you write a professional email to the HR manager requesting an update on my job application? My name is Priya Sharma and my registered email is priya.sharma@gmail.com. You can also mention that they can contact me on +91 98765 43210 if they need any additional information. Please make the email polite and professional.

Sentinel · 14:23:55

Scan: Email, Phone — MASKED · 7143ms

AI Assistant · 14:24:03

Ask me anything... (try including an email or Aadhaar to see blocking)

Press **Enter** to send · **Shift+Enter** for new line

Interception Inspector

Last scan · 14:42:12

VERDICT

BLOCKED

ENTITIES (2)

Gov ID (PAN)0.98BLOCK

Aadhaar0.99BLOCK

TOTAL LATENCY

65ms

CLEANED (SENT TO AI)

I need to write a message to customer support regarding my KYC verification. My name is Priya Sharma and my PAN number is [60V_ID]. My Aadhaar number is [AADHAAR]. Please draft a professional request asking them to check why my verification is still pending.

BLOCKED REASONS

Gov ID (PAN) (BLOCK), Aadhaar (BLOCK)

User Interface

Sentinel

AI GOVERNANCE LAYER

MONITOR

SOC · Live Flow

Live Overview

Event Stream2500

Audit Logs

Playground

AI Chat

CONFIGURE

Detection Rules

Connected Agents

Firewall active

5 agents · 4137 req/min

AI Chat — Safe Conversations

sentinel-prod-01 · us-east

STREAMING

presidio: ready

SR

gemin: not configured

AI Chat

Chat with AI through Sentinel's interception layer — safe prompts reach the LLM, PII/toxic prompts are blocked

You · 14:23:55

Hi, can you write a professional email to the HR manager requesting an update on my job application? My name is Priya Sharma and my registered email is priya.sharma@gmail.com. You can also mention that they can contact me on +91 98765 43210 if they need any additional information. Please make the email polite and professional.

Sentinel · 14:23:55

Scan: Email, Phone — MASKED · 7143ms

AI Assistant · 14:24:03

Here's a professional email to the HR manager requesting an update on your job application: Subject: Request for Update on Job Application Dear [HR Manager's Name], I hope this email finds you well. I am writing to express my interest in the [Job Title] position that I applied for on [Application Date]. I am still eager to join the team and contribute my skills and experience to the company. I was wondering if you could provide me with an update on the current status of my application. I understand that hiring processes can take time, and I would appreciate any information you can share regarding the progress of my application. If there is any additional information or documentation required from me, please do not hesitate to contact me. You can reach me at [PHONE] or via email at [[EMAIL]](mailto:[EMAIL]). I would be grateful if you could let me know if there are any further steps I need to take or if there is any other information you require from me. Thank you for your time and consideration, and I look forward to hearing from you soon. Best regards, Priya Sharma

You · 14:28:39

Ask me anything... (try including an email or Aadhaar to see blocking)

Press

Enter

 to send ·

Shift+Enter

 for new line

Interception Inspector

Last scan · 14:42:12

VERDICT

BLOCKED

ENTITIES (2)

Gov ID (PAN)0.98BLOCK

Aadhaar0.99BLOCK

TOTAL LATENCY

65ms

CLEANED (SENT TO AI)

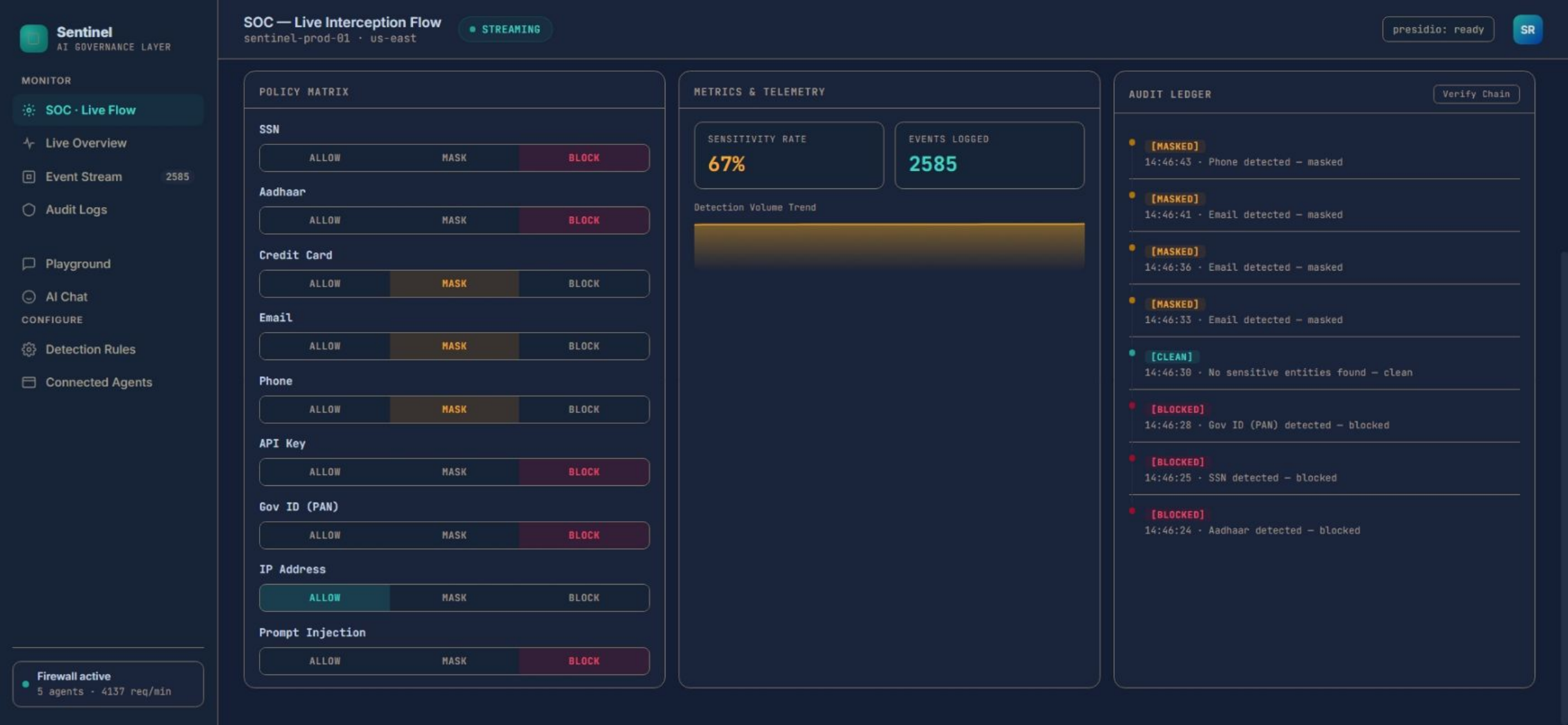
I need to write a message to customer support regarding my KYC verification. My name is Priya Sharma and my PAN number is [GOV_ID]. My Aadhaar number is [AADHAAR]. Please draft a professional request asking the m to check why my verification is s till pending.

BLOCKED REASONS

Gov ID (PAN) (BLOCK), Aadhaar (BLOCK)

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User Interface



User Interface

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Loading personal and system profiles took 2533ms.
PS C:\Users\sabar> Invoke-RestMethod -Uri "http://127.0.0.1:8000/api/inspect" -Method Post -ContentType "application/json" -Body '{"message": "My Aadhaar is 1234 5678 9012 and email is john@example.com", "agent_id": "agent-01", "session_id": "demo-001"}'

verdict          : MASKED
cleaned_message  : My [ORGANIZATION] is [DATE_TIME] and email is [EMAIL_ADDRESS]
entities         : {@{type=Email; action=MASK; confidence=1.0; original=john***}, @{{type=Organization; action=MASK; confidence=0.85; original=Aadh***}, @{{type=Date_Time; action=MASK; confidence=0.85; original=1234***}, @{{type=Url; action=MASK; confidence=0.5; original=exam***}}}
risk_score       : 70
latency_ms       : 52.46
audit_hash       : 138893eâ|caf
timestamp        : 16:08:28
agent_id         : agent-01
session_id       : demo-001

PS C:\Users\sabar> |
```

SOURCE CODE

[SabarishR08/ai-safety-governance-layer: Real-time PII detection, masking, and audit-log firewall for AI/agent systems — PEC Techathon 4.0](#)

FEATURE ENHANCEMENT

- Tool-call / tool-response scanning — extend interception to MCP tool metadata and outputs, closing the same trust gap exploited in tool poisoning attacks
- Out-of-band exfiltration detection (encoded data in rendered links/images)
- Local/on-prem LLM inference (Ollama) for zero-trust environments, no third-party API dependency
- Custom Presidio recognizers with checksum validation for Aadhaar/PAN (cuts false positives)
- Multi-language PII/toxicity coverage (Tamil, Hinglish) — relevant given your audience

Thank you